

Questions

This involves the equation

$$\begin{aligned}\partial_t u &= \partial_x^2 u + \sigma(u) \dot{W} \\ u(0, x) &\geq 0\end{aligned}\tag{0.1}$$

Let $\sigma(0) = 0$ and $0 \leq \sigma(u) < u^\gamma$, with $\gamma < 3/2$.

1 Blowup for colored noise

What if we replace $\dot{W}$ by noise that is colored in space but white in time. Note: Pavel Bezdek, Davar's student, did this for white noise and fractional Laplacian as the operator.

Davar said Pavel had results for colored noise, but didn't write it up.

2 Nonblowup on noncompact domains

Prove the non-blowup results on noncompact domains. We would need the following analysis.

2.1 Analysis of $N(t, x)$

Assume that $u_0 \in L^1$. We need to analyze $u(t, x)$ as $x \rightarrow \infty$ in order to study

$$\sup_{0 \leq t \leq T} \sup_{|x-a| \leq M} |N(t, x)|$$

2.2 Growth of u as $x \rightarrow \pm\infty$

We would need to study this to get information about $N(t, x)$ for large values of x .

3 Wave equation question

We might be able to prove non-blowup if we knew that:

Given two martingales $M_t^i; i = 1, 2$ which are continuous and given that

$$X_t := tM_t^{(1)} + M_t^{(2)} \geq 0$$

can we conclude that X_t is a.s. bounded on $[0, T]$.

4 Hitting Zero: Balancing the terms

$$\partial_t u = \partial_x^2 u + u^{-\alpha} + u^\gamma \dot{W}(t, x)$$

We want $\gamma < 1$. What are the conditions on α, γ so that u can or can't hit 0.

What about colored noise?